Δ_s: how far one person can move a release

DP protects the people in O's register from R, the only party that sees decrypted output

Who is protected

Neighbours D ~ D′: one matched person's (x_O, y) is replaced by any other value within the bounds. Their R row $\dot{x}$ and the match set I stay fixed.

- R already knows $\dot{x}$ and I (the linkage reveals I by design: outside DP).
- O sees only ciphertexts (encryption, not DP).

What release s contains

s	Stage	Entries released
1	stage 1	triu B, C
2	stage 2	triu B, C, c_R, c_O, y^Ty

A = $\dot{X}^T\dot{X}$ is never released: R computes it itself, and it is the same in both neighbours.

The sensitivity Δ_s

$$\Delta_s = \sup_{D \sim D'} \left\| z_s(D) - z_s(D') \right\|_2$$

With $\dot{x}$ fixed and $x = x_O$, a replacement $(x, y) \to (x', y')$ moves:

$$\begin{aligned} \operatorname{triu} B: &\quad \operatorname{triu}\left(xx^\top - x'x'^\top\right) \\ C: &\quad \dot{x}\,(x - x')^\top \\ c_R: &\quad \dot{x}\,(y - y') \\ c_O: &\quad x\,y - x'\,y' \\ y^\top y: &\quad y^2 - y'^2 \end{aligned}$$

Every entry gets the same noise, so the squared changes add up over the blocks of a release. Stage 1 has only the first two blocks, so $\Delta_G < \Delta$. Only the upper triangle of the symmetric B is released, so each off-diagonal change counts once.

Maximizing over the bounds, with (r, o, w) = (B_R^2, B_O^2, B_y^2):

$$\Delta_2^2 = \Delta^2 = \tfrac{1}{2}\,(2o + r + w)^2 + 2rw \qquad (|r - w| \le 2o)$$
$$\Delta_1^2 = \Delta_G^2 = \tfrac{1}{2}\,(2o + r)^2 \qquad (r \le 2o)$$

Toy: $B_R^2 = 3$, $B_O^2 = 2$, $B_y^2 = 1 \to \Delta = \sqrt{38} = 6.16$, $\Delta_G = \sqrt{24.5} = 4.95$.

Clipping is what makes Δ finite, and Δ does not depend on n: more records make the signal bigger, not the noise.

σ_s and ρ_s: one equation ties them to Δ_s

Δ_s is fixed by the bounds and ρ_s by the budget, so the noise σ_s follows

$$\rho_s = \frac{\Delta_s^2}{2\,\sigma_s^2} \qquad \Longleftrightarrow \qquad \sigma_s = \frac{\Delta_s}{\sqrt{2\rho_s}}$$

Δ_s sensitivity: how far one person can move release s
ρ_s zCDP cost: the share of the budget release s spends
σ_s noise s.d. added to every entry of release s

Why the equation holds

R sees $\hat{z}_s = z_s + N(0, \sigma_s^2\, I)$: independent noise with the same s.d. on every entry. For neighbours D ~ D′ and any order a > 1, the Rényi divergence is

$$D_a\big(\mathcal{N}(z_s(D), \sigma_s^2 I) \,\|\, \mathcal{N}(z_s(D'), \sigma_s^2 I)\big) = \frac{a\,\|z_s(D) - z_s(D')\|_2^2}{2\sigma_s^2} \le a\,\frac{\Delta_s^2}{2\sigma_s^2}$$

ρ-zCDP means this is at most aρ for every a > 1, so release s costs exactly $\rho_s = \Delta_s^2/(2\sigma_s^2)$. Costs add: $\rho_{run} = \rho_1 + \rho_2$. They add across runs too, and O's ledger caps the total.

From the run's (ε, δ) to the noise

1 Convert (ε, δ) to a zCDP budget

$$\sqrt{\rho_{\text{run}}} \gtrsim \sqrt{L + \varepsilon} - \sqrt{L}, \qquad L = \ln(1/\delta)$$

Tight conversion: $\rho_{run} = 0.01497$ at ε = 1, δ = 10^{-9}.

2 Split it between the two stages

$$\rho_1 = f_1\,\rho_{\text{run}}, \qquad \rho_2 = (1 - f_1)\,\rho_{\text{run}}, \qquad f_1 = 0.1$$

3 Set the noise

$$\sigma_1 = \Delta_G/\sqrt{2 f_1 \rho_{\text{run}}}$$
$$\sigma_2 = \Delta/\sqrt{2(1 - f_1)\rho_{\text{run}}}$$

$\sigma_1 \approx 2.4\sigma_2$ on purpose: stage 1 only sets the ridge, and its 10% share raises σ_2 by just 5%.

Toy example (ε = 1, δ = 10^{-9}): $\Delta_G = 4.95$, $\rho_1 = 0.0015$, $\sigma_1 = 90.45$; $\Delta = 6.16$, $\rho_2 = 0.0135$, $\sigma_2 = 37.55$. Everything after the releases (λ, the gate, $\tilde{\theta}$, V_{mech}, the intervals) uses only $\hat{z}_1$, $\hat{z}_2$, A and public numbers: post-processing, at no extra privacy cost.

Where the noise actually lives

O adds integer noise η to the ciphertext's coefficients; R decodes it as Gaussian noise on each entry

1 Statistic in the slots

Release s is a vector z_s with one entry per slot. One person moves it by at most Δ_s in $\ell 2$.

encode → ×Ω, τ^{-1}

2 Coefficients: O adds η

$$\eta \sim \mathcal{N}_{\mathbb{Z}}(0, \sigma_{c,s}^2)^N$$

$$\Delta_{\mathrm{enc},s} = \sqrt{2/N}\,\Omega\,\Delta_s + \sqrt{N}$$

$$\rho_s = \Delta_{\mathrm{enc},s}^2 \,/\, (2\sigma_{c,s}^2)$$

exact for the discrete Gaussian

decode → ÷Ω, τ

3 What R decodes

$$\sigma_s = \sigma_{c,s}\sqrt{N/2}\,/\,\Omega$$

Noise with s.d. σ_s on every released entry: the picture of the previous slide.

Why the slot picture is enough

$$\sigma_s = \frac{\Delta_s + N/(\sqrt{2}\,\Omega)}{\sqrt{2\rho_s}} \approx \frac{\Delta_s}{\sqrt{2\rho_s}}$$

With $N = 2^{14}$ and $\Omega \approx 2^{61.5}$ the extra term is below 10^{-14}. Sampling the discrete Gaussian exactly, with integers, makes the guarantee exact rather than approximate.

η also hides the CKKS error

$$\mathrm{SD}(\eta + e,\ \eta) \le \frac{\|e\|_2}{\sqrt{2\pi}\,\sigma_c} \le 2^{-\kappa} \quad \text{if} \quad \sigma_c \ge \frac{2^{\kappa} B_{\mathrm{tot}}}{\sqrt{2\pi}}$$

The error e depends on the data and on O's key share, so it must be hidden (κ = 40). Since $\sigma_{c,s} = \sigma_s \cdot \Omega \cdot \sqrt{(2/N)}$, O raises Ω until this holds:

$$\log_2 \Omega \approx \underbrace{\kappa}_{40} + \underbrace{\log_2 B_{\mathrm{tot}}}_{25.3} + \underbrace{\log_2\sqrt{N/2}}_{6.5} - \underbrace{\log_2\sqrt{2\pi}}_{1.3} - \underbrace{\log_2 \sigma_2}_{9.0} \approx 61.5$$

Price: the DP itself gets $\delta_{\mathrm{DP}} = \delta_{\mathrm{run}} - n_{\mathrm{rel}}(\delta_B + 2^{-\kappa})$.

How the σ's drive the statistics: σ_1 sets the radius $r_1 = \sigma_1(2\sqrt{p} + 2\sqrt{\ln(1/\alpha_L)})$ and so λ; σ_2 sets the ridge target $2\vartheta^*\sigma_2\sqrt{p}$, the gate and $V_{\mathrm{mech}} \propto \sigma_2^2$. Records needed for a given coefficient accuracy grow like $\sigma_2^2 \propto \Delta^2/\rho$.